

The Lopsided Diffusion

Three Asymmetries in How Generative AI Reaches the World

Audit synthesis

Padilla, Lam, Lambrecht & Hollenbeck (2025), *Working Paper* | Liu, Huang & Wang (2025), *World Bank / Rice*
Ide & Talamàs (2025), *Journal of Political Economy* 133(12):3762–3797

−41.8% | 24% / 0.7% | **individual at**
 z_{AI} always loses

small-site traffic | HIC vs. LIC penetration | theoretical wage

The aggregate adoption curve is rising. The distribution beneath it is not.

Padilla et al. (2025), Table 7; Liu et al. (2025), p. 22; Ide & Talamàs (2025), Proposition 5.

Three lenses on diffusion — same direction, three asymmetries

Lens	What is uneven	Quantified asymmetry
Behavior	What adopters substitute (Padilla 2025)	Long queries −37.2%, education −89.3%, social media ≈ 0
Geography	Who gets to adopt (Liu 2025)	HIC penetration 24%, LIC 0.7%; log-GDP elasticity 0.6–0.8
Theory	Where the gains land (Ide 2025)	Winners at the extremes; $w^*(z_{AI}) < w(z_{AI})$ always

The aggregates conceal three different distributions of who gains, who substitutes, and who is left out.

Lens 1 — Behavior

which sites and which queries get substituted

LLM adopters cut search by 21.7%, but the cut is concentrated in informational queries

Search type (Table 5)	ATT	Pre-mean	% change
All search	-7.08**	32.67	-21.7%
Google only	-5.57*	17.68	-31.5%
Question searches	-1.26***	3.78	-33.2%
Long (≥ 6 words)	-2.47***	6.63	-37.2%
Pure navigational	-0.24 n.s.	0.80	—

Effects emerge ~ 20 weeks post-adoption — a learning period before substitution. Staggered DiD (Callaway–Sant’Anna 2021); $N = 2,041$ adopters; Comscore desktop panel.

Padilla et al. (2025), Tables 4–5; ATT for weeks 20–47.

The substitution falls hardest on small sites and education — not on the social web

Outcome	ATT	Pre-mean	% change
Bottom-25% websites (Tab 7)	−482.3**	1,154.3	−41.8%
All education sites (Tab 9)	−151.5**	169.7	−89.3%
Stack Overflow (Tab 10)	−0.65*	0.94	−68.8%
Wikipedia / Reddit (Tab 10)	both n.s.	—	—
Display ads (Tab 11)	−122.2*	161.1	−72.9%

Top-500 sites unaffected (+349.0 n.s.). Long-tail informational supply gets cannibalized; social, navigational, and short-query traffic is spared.

Padilla et al. (2025), Tables 6–11.

Lens 2 — Geography

who is allowed onto the curve at all

ChatGPT traffic +113% YoY — but penetration ranges from 24% in HIC to 0.7% in LIC

Group	Apr 2024 (M)	Apr 2025 (M)	User growth	Penetration
HIC	1,142	2,545	+43%	24%
UMC	490	1,000	+42%	5.8%
LMC	422	816	+37%	4.7%
LIC	2.3	~ 10	small base	0.7%

Decomposition: $+113\% = (1+0.42)(1+0.50) - 1$ — extensive (+42% users) and intensive (+50% visits per user) margins both contribute. Authors (p. 26): “the widening divide is ... more about adoption itself.”

Liu, Huang & Wang (2025), Section 4; Semrush panel, Dec 2022–mid-2025.

Log GDP per capita predicts adoption with elasticity 0.6–0.8; Google has saturated

OLS coefficient	ChatGPT	Gemini	Claude	Google
log GDP per capita	0.62***	0.79***	0.63***	0.16*
% pop. on internet	0.031***	0.017	0.019**	−0.012**
R^2	0.67	0.55	0.61	0.12
N countries	153	118	131	164

Localization is real: Le Chat 69% of traffic from Europe; Doubao and Tencent Yuanbao > 90% domestic. Authors: “correlation, not causation” (p. 29).

Liu et al. (2025), Table 4 (p. 29); Figure 17.

Lens 3 — Theory

why the gains land at the extremes

Garicano hierarchies + compute: AI agents at exogenous knowledge $z_{AI} \in [0, 1)$

Object	Definition
$z \sim G$ on $[0, 1]$	Human knowledge; partition into W (workers), I (independents), S (solvers)
$h < h_0$	Solver time per question; admits two-layer firms
m (compute)	Abundant; 1 unit $\rightarrow$ 1 AI agent of fixed knowledge z_{AI}
Autonomous AI	Agent acts as worker AND solver; nonautonomous = copilot only

Prop 2. $r^* = z_{AI}$. If $z_{AI} \in \text{int } W$, basic AI; if $z_{AI} \in \text{int } S$, advanced AI — displacement direction flips.

Ide & Talamàs (2025), *JPE*; pure theory; Props. 1–2.

Proposition 5: winners at the extremes; the individual at z_{AI} always loses

Set	When non-empty
$T = (z_t, 1]$ (top winners)	Always non-empty under $h < h_0, z_{AI} < 1$
$B = [0, z_b)$ (bottom winners)	Non-empty iff $z_{AI} > \bar{z}_{AI} \in \text{int } W$
$\{z_{AI}\}$ (pivot individual)	$w^*(z_{AI}) < w(z_{AI})$ <i>always</i>

The shape of the welfare curve is a U with a cliff at z_{AI} . The least knowledgeable gain only when AI is capable enough; the most knowledgeable always gain; the human who does exactly what the AI does is uniformly displaced.

Ide & Talamàs (2025), Proposition 5, p. 3788.

Output ranking: autonomous AI > nonautonomous > no AI — regulation trades output for inequality

Regime	Best for least-knowledgeable	Best for most-knowledgeable
Autonomous AI	only if $z_{AI} > \bar{z}_{AI}$	always
Nonautonomous AI	always, if $z_{AI} > w(0)$	dominated by autonomous
No AI	—	—

Reconciles micro evidence: copilot RCTs (Brynjolfsson–Li–Raymond 2025) map to nonautonomous → help the bottom. Coworker deployments (Klarna; Alekseeva 2024) map to basic autonomous → displace the bottom. Same AI is “basic” in HIC and “advanced” in LIC — direction flips by economy.

Ide & Talamàs (2025), Prop. 6; §VII, pp. 3791–3796.

Devil's advocate: aren't these three lenses measuring different things at different times?

The strongest objection

Padilla is desktop-only, US-skewed, $N=2,041$. Liu is web-traffic only — no API, mobile, or enterprise. Ide is pure theory with $z_{AI} < 1$. Maybe the asymmetries are artifacts of measurement.

Reply. The lenses converge because they measure complementary slices of one substitution:

- Padilla's bottom-25% sites and education = Ide's $z < z_{AI}$ long-tail informational supply.
- Liu's 0.6–0.8 log-GDP elasticity says z_{AI} 's relative position is country-specific.
- Ide's pivot at z_{AI} explains why Stack Overflow falls -68.8% but Reddit does not.

The asymmetry is structural, not measurement noise.

The diffusion is lopsided in three layered ways — they compound, they do not cancel

Layer	Asymmetry	Compound effect
Behavior	Long-tail informational supply substituted, social web spared	Site economics flip for the bottom 4.5M websites
Geography	HIC adopt; LIC do not	Where z_{AI} sits in the local skill distribution differs by country
Theory	Welfare U-shape with cliff at z_{AI}	The cliff lands on different occupations in HIC vs. LIC

The same AI cannot be expected to produce the same outcomes across users, countries, or skill bands. Aggregation hides the cliff.

The curve is rising.

Beneath it, the cliff is rising too.